

# ClosetScan Interview Preparation

AI & AR/VR Innovation Intern | Second-round technical demonstration

**Confirmed: Tuesday, July 21, 2026 · 9:00 AM Eastern (EST)**

## 1. Logistics

**Candidate:** Latha Prabhakar

**Contact:** jimkliew

**Email:** lathaprabhakar243@gmail.com

**Repo:** <https://github.com/latha0624/ClosetScan>

**Device:** iPhone 16 Pro (LiDAR)

**Subject line:** AI & AR/VR Innovation Intern

**Format:** ~20 minute live technical demo + Q&A;

## 2. One-sentence pitch

ClosetScan uses Apple RoomPlan and LiDAR to scan a closet, digitally hide its contents as an empty shell, and report width, depth, and height to the nearest 1/16 inch — with a tape-measure validation workflow.

## 3. Night-before checklist

- Confirmation email sent to jimkliew (9:00 AM EST Tue Jul 21)
- Repo link shared: <https://github.com/latha0624/ClosetScan>
- ClosetScan installed on iPhone 16 Pro and opens cleanly
- If Sideloadly: re-signed within last 7 days
- Camera / LiDAR / World Sensing permissions granted
- Phone charged  $\geq 50\%$ ; Low Power Mode off
- Closet ready: door fully open, clear path, decent light
- Steel tape measure ready for validation
- $\geq 3$  practice scans done; validation history populated
- Backup: last successful scan still viewable from Home
- Quiet space + meeting link tested (if remote)

## 4. Morning-of (arrive 15–20 min early)

1. Unlock phone; open ClosetScan once to warm LiDAR / permissions.
2. Place tape measure where you can reach it during the demo.
3. Open GitHub repo on a second screen if you want to show code.
4. Silence notifications.
5. Keep this prep PDF nearby for Q&A; only if needed.

## 5. 20-minute live demo script

| Time        | Beat         | What to say / do                                                                                        |
|-------------|--------------|---------------------------------------------------------------------------------------------------------|
| 0:00–0:30   | Home         | ClosetScan — RoomPlan + LiDAR + RealityKit. Goal: empty shell + dimensions to 1/16", validated vs tape. |
| 0:30–3:30   | Live scan    | Open closet door fully. Scan slowly; cover walls, floor corners, and upper extent.                      |
| 3:30–5:00   | Empty shell  | Toggle Show contents On → Off. Surfaces only = digital empty shell; objects omitted on purpose.         |
| 5:00–6:00   | Dimensions   | Read $W \times D \times H$ in fractional inches. Mention meters + decimal + 1/16" display.              |
| 6:00–8:00   | Validation   | Enter steel-tape W/D/H. Show absolute error vs 1/16" target and prior trials.                           |
| 8:00–16:00  | Architecture | Flow, content removal, dimension math, accuracy honesty, limitations.                                   |
| 16:00–20:00 | Q&A;         | Short, concrete answers. Offer to dig into code if asked.                                               |

### Demo tips

- Narrate while scanning: “covering the back wall... corners... upper shelf line.”
- If scan quality looks weak, do a second pass rather than apologizing at length.
- Always land on the accuracy honesty line — do not oversell  $\pm 1/16$ " sensor accuracy.

## 6. Architecture talking points

Home → RoomCaptureView (scan) → CapturedRoom

- EmptyClosetViewer (surfaces ± objects)
- ClosetDimensions (AABB → fractional inches)
- ValidationStore (tape vs app trials)

### Key files

#### App state — AppModel.swift

Phase machine: home → scanning → results → validation

#### Scan — RoomCaptureController.swift, ScanView.swift

RoomCaptureView wraps RoomPlan; captures parametric room

#### Math — ClosetDimensions.swift

Surface corners → world AABB → W/D/H → nearest 1/16"

#### 3D UI — EmptyClosetViewer.swift, ResultsView.swift

RealityKit dollhouse; contents toggle

#### Accuracy — ValidationView.swift, ValidationStore.swift

Persist trials; absolute error vs tape

### Content removal (expected question)

RoomPlan returns **Surfaces** (walls, floors, doors/windows/openings = empty envelope) and **Objects** (storage, furniture). Digital removal = **intentional omission**: build meshes from surfaces only unless the contents toggle is on. No generative inpainting. Reproducible under interview lighting.

### Dimension math (expected question)

- Transform each surface's local corner extents into world space.
- Axis-aligned min/max → extent.
- Y → height; larger horizontal axis → width, smaller → depth.

- Meters → inches ( $\times 39.37007874015748$ ), round to nearest 1/16", reduce fraction.

## 7. Accuracy — say this exactly

**Display resolution:** 1/16" ( $\approx 1.5875$  mm). **Sensor reality:** LiDAR + RoomPlan is generally **centimeter-class**. Do **NOT** claim guaranteed  $\pm 1/16''$  sensor accuracy. Do claim: *we measure, report to 1/16'', and validate against tape.*

### Validation protocol

1. Open door fully; scan slowly covering all walls and upper extent.
  2. Record app W / D / H.
  3. Measure same axes with steel tape (same reference edges).
  4. Enter tape values in Validate Accuracy.
  5. Repeat  $\geq 3$  independent scans.
  6. Report MAE and max absolute error per axis.
- 1/16" = 0.0625". Fill with your practice trials before the call:

| Trial | App W | Tape W | \Delta W | App D | Tape D | \Delta D | App H | Tape H | \Delta H | $\leq 1/16''$ ? |
|-------|-------|--------|----------|-------|--------|----------|-------|--------|----------|-----------------|
| 1     |       |        |          |       |        |          |       |        |          |                 |
| 2     |       |        |          |       |        |          |       |        |          |                 |
| 3     |       |        |          |       |        |          |       |        |          |                 |

## 8. Known limitations (own them)

- Requires LiDAR; non-Pro iPhones cannot run RoomPlan capture.
- Very narrow or cluttered closets may under-detect walls or merge surfaces.
- Clothes/hanging items may register as objects inconsistently; empty-shell still shows structure.
- AABB can slightly overestimate if door frames / openings expand the bounds.
- Width vs depth is normalized (larger horizontal first) for stable UI numbers.

## 9. Likely Q&A;

### Q: Why RoomPlan instead of raw ARKit mesh?

A: Parametric surfaces + objects give a clean empty-shell toggle and stable dimensioning without fighting a dense mesh. Faster path to a clear demo narrative.

### Q: How accurate is 1/16"?

A: Display precision, not guaranteed sensor precision. We validate against tape and report error honestly.

### Q: What would you build next?

A: Opening-aware clear span, multi-scan fusion, USDZ export/share sheet, ML clothing segmentation on the empty shell, or charts for validation history.

### Q: Why SwiftUI + RealityKit?

A: Native RoomPlan integration, fast UI for interview iteration, RealityKit for the dollhouse visualization.

### Q: Did you need a Mac?

A: No — free path via GitHub Actions IPA + Sideloadly on Windows (FREE\_INSTALL.md). Device is iPhone 16 Pro.

**Q: What broke during development?**

A: RoomPlan API details (no ceilings array — height from wall extents), RoomCaptureViewDelegate / NSCoder conformance for CI. Happy to walk through commits.

## 10. Closing line

**“Thanks for the time — happy to walk through any part of the scan pipeline, the dimension math, or the validation results in more detail.”**

## 11. Confirmation email draft (send tonight)

**Subject:** AI & AR/VR Innovation Intern — confirming Tuesday 9:00 AM EST

Hi Jim,

I am writing to confirm our second-round technical interview for the AI & AR/VR Innovation Intern role.

**Confirmed time:** Tuesday, July 21, 2026 at **9:00 AM Eastern (EST)**.

I will demonstrate ClosetScan, an iPhone LiDAR + RoomPlan demo that scans a closet, digitally hides contents as an empty shell, and reports dimensions — to the nearest 1/16", with tape-measure validation.

Repo: <https://github.com/latha0624/ClosetScan>

Please let me know if this time still works on your side, or if you prefer a different slot. Looking forward to the conversation.

Best regards,  
Latha Prabhakar  
lathaprabhakar243@gmail.com